

Vesta: The Takeout Question

*Providence, Rhode Island.*¹

The line starts forming outside Vesta at about a quarter past six, and on a good Friday it does not really stop until nine. June Ellery has managed the room for four years and she still cannot walk past the front window at seven o'clock without counting. Eighteen tables. Fifty seats. No reservations, ever, which was Marco's condition when they opened and which has turned out to be the single best decision anybody made, because the line is the advertising.

What June has never had is a way to answer a question about the line other than by standing in it.

The question in front of her arrived three weeks ago in the form of Owen Brandt, who put up a third of the money to open Vesta and who has spent the last year watching every other restaurant on the block put its food in a paper bag.

"You are turning people away at the door four nights a week," Owen said. "I have watched you do it. Those people want your food. They do not need your table. Put the menu on the platform, take the orders, hand them a bag. It is the same kitchen, the same food, and it is demand you are currently throwing into the street."

Marco Devlin, who runs the kitchen and owns the other third, put his knife down when he heard this, which June had learned to read as a bad sign.

"It is not the same kitchen. There is one line. There are three cooks on it. Every bag that goes out the front door is a plate that did not go to table nine, and the people at table nine are sitting in my dining room watching their food not arrive. You want me to make my best customers wait so a stranger can eat my food out of a box on his couch."

"They are all customers, Marco."

"They are not all paying for a table."

June's job, by the end of the month, is to tell the three of them whether Vesta should take takeout orders on Friday nights, and to be able to say why.

What everybody agrees on

The physical facts are not in dispute, and are in Exhibit 1. Doors open at 4:30. The last party is seated at 9:30, and anyone still in line at 9:30 gets seated. Eleven tables seat two, seven seat four. Four servers work the floor and three cooks work the line, and all seven of them are paid for the night whether Vesta serves ninety covers or two hundred. So is the rent, so is June, and so is the dishwasher. What varies with the night is food and beverage cost, which runs a little over thirty cents on the dollar, and that is very nearly the whole of it.

Tessa Kimura works the door, and Tessa quotes waits by a rule she has never written down but which she recited when June asked: count the parties ahead of you who need the same kind of table you do, add one, multiply by seventy-two minutes, divide by the number of tables in the room that seat your size, and round to the nearest five. If there are more of those tables sitting empty than there are parties ahead of you, the wait is zero and she walks you in.

¹Teaching case prepared for classroom discussion. Vesta, the delivery platform and all individuals are fictional; all quantitative figures are illustrative values constructed for this exercise. **Note to students:** this case is deliberately not a complete specification. Where the facts underdetermine the model, that is the assignment, not an error: identify the gap, choose a defensible reading, and record it in your Assumption Register.

What nobody agrees on

Three weeks of arguing has produced four positions and no decision.

Owen wants the volume. “Thirty-five orders a night at forty-three dollars is fifteen hundred dollars of food we are not selling. I do not need it to be elegant.”

Marco wants his dining room protected. “Dine-in first. Always. If a bag has to wait, the bag waits.” Pressed on what happens when the bag has waited forty minutes and the platform is calling, he said that was not his problem. Pressed again, he admitted he did not actually know what the expediter would do at eight o’clock on a Friday, because nobody had ever told the expediter anything.

Tessa wants the line to stay honest. “I quote thirty-five minutes and people believe me. If I start quoting thirty-five and delivering fifty, they stop believing me, and then they stop coming.” She also pointed out, with some feeling, that the four people on the floor work for tips, that a table tips twenty percent, and that a bag on a shelf tips almost nothing. “You are asking the kitchen to do the same work for a third of the money, and you are asking my servers to hand out the bags for free.”

And June is not sure the question is even the one being asked. What she keeps coming back to is the eighteen parties who walked away from the door last Friday. She knows roughly what happens to them, because Vesta ran a loyalty programme for two years before it got too complicated to administer, and the numbers are in Exhibit 5: about a third of the people turned away never come back at all, and the ones who do come back come back about four times a year. Whether that belongs in the arithmetic, and at what number, is the thing she cannot get any of the other three to engage with.

What the platform is offering

The terms are in Exhibit 5. The platform forecasts about thirty-five orders on a Friday, mostly between six and eight; it takes twenty-eight percent of the ticket on the orders it brings, which it expects to be three in four; the rest phone in and pay nothing. Vesta quotes a ready time of twenty-five minutes when the order is accepted. Orders that run well past that get refunded, and the food gets thrown away. Restaurants that miss the quoted time too often get pushed down the rankings.

Owen, demonstrating the tablet, mentioned in passing that there is a toggle on the merchant dashboard that pauses new orders. “Every restaurant has it. Nobody ever touches it.”

What June has

June pulled three files off the point-of-sale system for last Friday, a completely ordinary Friday in November. They are in the data folder.

`data_door_log.csv` is the host stand’s record: one row per party that came to the door, with the time it arrived, how many people, the wait Tessa quoted, whether it stayed or left, and for the ones that stayed, which table and when they left.

`data_kitchen_tickets.csv` is the ticket printer’s record: one row per ticket, with the time it was fired to the line and the time the finished plate was bumped. The printer is old. It stamps those two times and nothing in between, so it does not record when a cook actually picked the ticket up.

`data_checks.csv` is one row per table: covers, what they spent, what they tipped, and how long they sat.

`data_takeout_forecast.csv` is the platform’s projected order volume by half hour.

Summary figures from all four are in Exhibits 2 through 5.

The charge

June has to bring Marco and Owen a recommendation, and she has decided that “it depends” is not one. She needs to say whether Vesta should take takeout orders on Friday nights; if so, under what operating rules; and she needs to be able to say how confident she is, because Owen will ask, and because she has already watched him win an argument with a single good night.

Exhibits

All figures are illustrative values constructed for this exercise.

Exhibit 1. The room, the crew and the clock

Tables for two	11
Tables for four	7
Seats	50
Servers on a Friday	4
Line cooks on a Friday	3
Doors open	4:30 pm
Last party seated	9:30 pm
Minutes to clear and reset a table	4

Vesta takes no reservations. Anyone still in line at 9:30 pm is seated.

Exhibit 2. Last Friday at the door

Parties that came to the door	90
Parties seated	72
Parties who looked at the quoted wait and left	17
Parties who joined the line and then gave up	1
Covers served	186

Parties of 1	7
Parties of 2	39
Parties of 3	15
Parties of 4	29

Median quoted wait, parties who stayed	0 min
Median quoted wait, parties who left	35 min
Longest quoted wait of the night	60 min

The full row-by-row log is in `data.door.log.csv`: one line per party, with the time it arrived, its size, the wait Tessa quoted it, whether it stayed, and if it stayed, which table it took and when it left.

Exhibit 3. Last Friday in the kitchen

Tickets fired	72
Covers on those tickets	186
Median ticket turnaround, fired to bumped	11.2 min
Ninetieth percentile ticket turnaround	18.2 min

Covers on a typical recent Friday	166
Cook-hours on the floor (3 cooks, 4:30 pm to 10:30 pm)	18
Of those, hours the expo tablet logs the line as actively cooking, averaged over recent Fridays	11.6

Ticket by ticket in `data.kitchen.tickets.csv`. Vesta's ticket printer stamps two times: when the ticket is fired to the line and when the finished plate is bumped. It does not record when a cook actually picked the ticket up, so turnaround and cooking are not the same thing.

Exhibit 4. What a table is worth

Party size	Tables last Friday	Median check (\$)	Median minutes at the table
1	7	34	43
2	36	66	58
3	9	82	78
4	20	125	68
All parties	72	77	64

Tips averaged 20.0 percent of the check and go to the floor and the kitchen, not to the house. Food and beverage cost runs about 30 percent of sales. Every other cost of a Friday night, the rent, the salaried staff, the four servers and the three cooks, is committed before the doors open. Check by check in `data_checks.csv`.

Exhibit 5. The takeout proposal

Hours takeout would be offered	5:00 pm to 9:00 pm
Platform's forecast of orders on a Friday	40
Mean spend per takeout order	\$43
Share of orders the platform expects to bring	75%
Platform commission on those orders	28%
Packaging per order	\$1.55
Ready time quoted to the guest	25 min
Tips on takeout orders	about 8%
Parties turned away who never come back (Vesta's own loyalty data)	30%
Visits a regular makes in a year	4

The forecast by half hour is in `data_takeout_forecast.csv`. The platform demotes a restaurant in its rankings if too many orders run past the quoted ready time, and an order that runs far past it is refunded, with the food thrown away.

Deliverables. (i) A Monte Carlo simulation of a Friday night at Vesta, built so that a reader could re-run it, together with a recommendation on the takeout proposal and on the operating rules that go with it; (ii) a two-page memo to Marco, Owen and Tessa, answering the question on each of their yardsticks, and stating how sure you are and why; (iii) a completed **Assumption Register** documenting every point at which this case underdetermines the model, the readings you considered, the one you chose, and what the alternative reading would have done to your answer.

A note on method. Your answer is a random variable. Report it like one.